

Select the correct answer using the code given below:

Code:

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

U.P. R.O./A.R.O. (Mains) 2021

Ans. (b)

Indonesia is made up of more than 17,000 islands. Indonesia is a volcanically active country, containing numerous major volcanoes. Indonesia has near about 150 total volcanoes in of which about 80 are active volcanoes.

27. The famous Volcanic mountain 'Karakatoa' is situated in which of the following countries?

- (a) Italy
- (b) Indonesia
- (c) United States of America
- (d) Japan

U.P. R.O./A.R.O. (Mains) 2016

Ans. (b)

Karakatoa (Krakatau) volcano is in the Sunda Strait between Java and Sumatra island in Indonesia.

28. Match List - I with List - II and select the correct answer using the codes given below the lists :

List - I (Volcanic Mountains)	List - II (Country)
A. Mount Rainier	1. Italy
B. Mount Etna	2. Mexico
C. Mount Paricutin	3. Philippines
D. Mount Apo	4. U.S.A.

Codes :

A	B	C	D
(a) 4	2	1	3
(b) 4	1	2	3
(c) 2	1	4	3
(d) 4	3	2	1

U.P.P.C.S. (Pre) 2021

Ans. (b)

The correct match is as follows -
 Mount Rainier U.S.A.
 Mount Etna Italy
 Mount Paricutin Mexico
 Mount Apo Philippines

The Earthquake

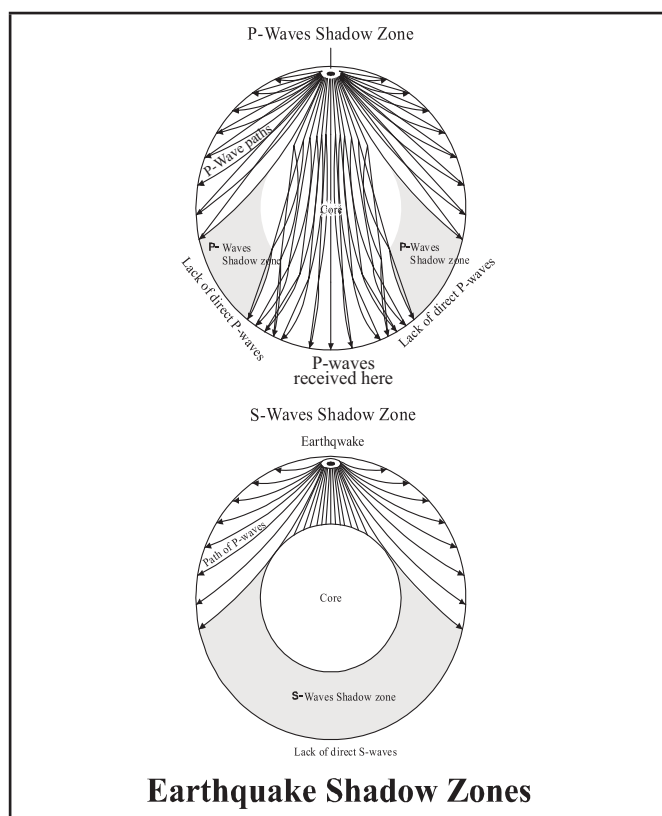
*An earthquake is the shaking of the surface of the Earth, resulting from the sudden release of energy in the earth's lithosphere that creates seismic waves. An earthquake's point of initial rupture is called its **seismic focus** or **hypocenter**. The epicenter is the point at ground level directly above the hypocenter. *The epicenter is the point where the waves of an earthquake are first observed. All natural earthquakes occur in the lithosphere. There are three basic types of seismic waves – (1) **Primary Waves (P-wave)** – They are similar to sound waves. They are also known as longitudinal waves. P-waves travel faster than any other waves. Their intensity of travel depends upon the density of the rock they encounter. Since they are the fastest traveling seismic wave, they are the first signal from an earthquake to arrive at any affected location or at a seismograph. **P-waves** may be transmitted through gases, liquids and solids.

(2) **Secondary Waves (S-waves)** – They are similar to light waves. They are also known as **transverse waves**. These waves arrive after the primary waves and travel slower than the P-waves. The S-waves can travel only through solid materials.

(3) **Surface Waves** – These waves travel slower than the other two waves and can travel only through the surface of the earth. Surface waves travel around the globe before reaching the epicenter. Surface waves are the most damaging of all the waves. Surface waves are of two types (1) Love waves (2) Rayleigh waves. In these two waves Love waves have higher speed.

*The magnitude of an earthquake is the indication of the energy released. *The magnitude of the earthquake is measured on the **Richter scale** and intensity on **Mercalli scale**. The Richter scale was invented in the year 1935 by Charles Francis Richter. The Richter scale is a logarithmic scale whose base is 10. Each step represents a ten-fold increase in amplitude on the seismogram, whereas in terms of energy, each whole number increase corresponds to an increase of about 32 times (31.6) the amount of energy released. Every point on the scale (earthquake waves) denotes an energy release, equivalent to 6 ounces of TNT (Trinitrotoluene).

***Seismometer** is used to measure earthquake waves. The term tsunami is borrowed from the Japanese meaning harbor wave. *Tsu means **Harbor** and **nami** means **waves**. Tsunami is caused by earthquakes, volcanic eruptions and other underwater explosions and their destructive power can be enormous affecting entire ocean basins.



*The **Fukushima Daiichi Nuclear disaster** was an energy accident at the Fukushima Daiichi Nuclear Power Plant initiated primarily by a tsunami following an earthquake on 11 March 2011. It led to a nuclear meltdown and the release of radioactivity in the area. *The **Ring of Fire** is a major area in the basin of the Pacific Ocean where many earthquakes and volcanic eruptions occur. The Ring of Fire is sometimes called Circum Pacific belt. About 90% of the world's earthquakes occur along the Ring of Fire.

*There are four main conditions for the occurrence of earthquakes in this region – (1) **Point of interaction between sea and land** (2) **Young fold mountainous region** (3) **Volcanic Activity** (4) **Plate tectonics** – movement and collision of plates. *The region consists of Chile, California, Alaska, Japan, Philippines, New Zealand etc. *The Northwestern part of the Indian Subcontinent is an earthquake prone area. This is because it is the convergence zone of Indian and Eurasian tectonic plates. *The earthquake-prone regions of India are included in the Mid-Continental Belt. The Belt starts from Cape Verde and includes Atlantic Ocean, Mediterranean Sea, and young fold mountain ranges like the Alps, Caucasus and Himalayas. The Mid-Atlantic Belt is a mid ocean belt.

1. Based on following statements choose the correct option:

Statement - I : Theory of Plate tectonics was given by H.H. Hess in 1967

Statement - II : Theory of Plate Tectonics is based on Theory of Sea floor Spreading

- (a) Statement - I and Statement - II both are true
(b) Statement - I and Statement - II both are false
(c) Statement - I is true but Statement - II is false
(d) Statement - I is false but Statement - II is true

Chhattisgarh P.C.S. (Pre) 2021

Ans. (d)

The concept of plate tectonics was formulated during the late 1960s. Wilson, Morgan, Parker, McKenzie, Harry Hess etc. contributed to the development of this theory. The theory of plate tectonics is based on the ocean (sea) floor spreading theory.

2. During an earthquake, which type of waves get generated?

- (a) B.S.L. (b) A.B.L.
(c) R.S.L. (d) P.S.L.
(e) F.S.L.

Chhattisgarh P.C.S (Pre) 2013

Ans. (d)

The seismic waves are generated in three forms. (1) P- (Primary) Waves – these waves travel fastest. (2) S - (Secondary) Waves- these waves cause particles to oscillate. Both P and S waves are called 'Body waves' because they can travel through the interior of the Earth. (3) Surface Waves – these waves arrive after P and S waves and are confined to the surface of the Earth during an earthquake. There are two types of surface waves (1) Love waves and (2) Rayleigh waves.

3. Read the following statement related to 'Earthquake shadow zone' and choose the correct option:

Statement I : The shadow zone of one earthquake is totally different from the shadow zone of another earthquake

Statement II : Seismometers record both 'P' and 'S' waves at any distance out of 105° from earthquake's epicentre

- (a) Statement I and Statement II both are true
(b) Statement I is true, but Statement II is false
(c) Statement I is false, but Statement II is true
(d) Statement I and Statement II both are false

Chhattisgarh P.C.S. (Pre) 2020

Ans. (b)

Earthquake waves get recorded in seismographs. However, there exist some specific areas where the waves are not reported. Such a zone is called the 'Shadowzone'. The study of different events reveals that for each earthquake, there exists an altogether different shadow zone. P waves don't arrive between 105° (103° - 105°) and 140° due to refraction at the mantle-core boundary, however S waves don't reappear beyond an angular distance of $\sim 130^{\circ}$ as they are stopped by the liquid outer core.

4. The 'Ring of Fire' is associated with

1. Earthquake
2. Volcanoes
3. Pacific Ocean
4. Forest Fires

Choose your answer with the help of codes given below :

Code :

- | | |
|----------------|-------------------|
| (a) 1, 2 and 3 | (b) 2 and 3 |
| (c) 2 and 4 | (d) 1, 2, 3 and 4 |

U.P.P.C.S. (Pre) 2008

Ans. (a)

'Ring of Fire' is a region in the Pacific Ocean influenced by volcanic activities and earthquakes. Approximately 90% of the world's earthquakes occur along the 'Ring of Fire'. The region of 'Ring of Fire' includes the vast seismic zone of Chile, California, Alaska, Japan, Philippines, New Zealand etc.

5. Consider the following statements :

1. The Richter scale is a logarithmic scale and so an increase of 1 magnitude unit represents a factor of 10 in amplitude.
2. Each integer reading of the Richter scale has an energy 100 times that the previous integer reading.

Which of the statement(s) given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

I.A.S. (Pre) 2006

Ans. (a)

The magnitude of the earthquake is measured on the Richter scale and intensity on the Mercalli scale. In 1935, seismologist Charles Francis Richter of the California Institute of Technology developed the Richter magnitude scale. It is based on a logarithmic scale whose base is 10. Each number on the seismogram represents 10 times amplitude and 32 times energy release. A seismic wave of magnitude 1 releases the energy of 6 ounces of TNT. Hence statement (1) is correct while statement (2) is wrong.

6. Consider the following statement about earthquake –

1. Intensity of earthquake is measured on Mercalli Scale.
2. Earthquake's magnitude is the measurement of energy released.
3. Magnitude of earthquake depends directly upon Amplitude of an earthquake waves.
4. Every integer on Richter Scale shows a 100 times increase in energy released

Which of the statements are true?

- | | |
|----------------|----------------|
| (a) 1, 2 and 3 | (b) 2, 3 and 4 |
| (c) 1 and 4 | (d) 1 and 3 |

I.A.S. (Pre) 2001

Ans. (a)

The intensity of an earthquake is measured by the Mercalli scale. Hence, statement (1) is correct. The magnitude of an earthquake is the measurement of energy released, thus statement (2) is correct. The Richter scale has a shaking amplitude 10 times that of the previous integer and shows an increase of 32 times in energy release. Energy equal to 6 ounces of TNT is released by an earthquake of a magnitude of 1. Hence statement (3) is correct but statement (4) is false.

7. Richter Scale is used to measure –

- | |
|-----------------------------|
| (a) Intensity of earthquake |
| (b) Depth of Ocean |
| (c) Velocity of Spacecraft |
| (d) Height of a building |

Uttarakhand P.C.S. (Mains) 2002

Ans. (a)

Though the Richter scale is not used to measure the intensity of earthquakes (Mercalli scale measures it), here option (a) is assumed to be correct. This is because option (a) is closest to the question asked.

8. The intensity of earthquakes is measured

- | | |
|--------------------------|-------------------------|
| (a) On the Richter scale | (b) On the Kelvin scale |
| (c) In decibel | (d) In Pascal |

U.P. Lower Sub. (Pre) 2015

Ans. (a)

See the explanation of the above question.

9. Seismometer measures –

- | | |
|----------------|-----------------------|
| (a) Heartbeat | (b) Height of trees |
| (c) Earthquake | (d) None of the above |

Uttarakhand Lower (Sub.) (Pre) 2010

Ans. (c)

The seismometer is an instrument to record the magnitude of an earthquake. A crescograph is used to measure the growth of plants, trees and cardiogram is an instrument to measure heartbeat.

10. Which of the following instruments is used to record seismic waves?

- (a) Seismogram
- (b) Seismograph
- (c) Seismoscope
- (d) Seismometer

U.P. P.C.S. (Mains) 2017

Ans. (b)

A seismograph is an instrument used to detect and record earthquakes. The seismometer is the internal part of the seismograph, which may be a pendulum or a weight mounted on a spring. However, it is also commonly used as a synonym for seismograph. A seismogram is a recording of an earthquake at a specific location on a seismograph instrument. Hence seismograph usually refers to the seismometer and its recording device as a single unit.

11. Tsunami word belongs to which of the following languages ?

- (a) Arabic
- (b) Japanese
- (c) Hebrew
- (d) Latin

Uttarakhand P.C.S. (Pre) 2005

Ans. (b)

Tsunami is a Japanese word "tsu" which means 'harbor' and "nami" means 'wave'. Thus tsunami means 'harbor wave' arising as a result of the earthquake. These ruptured high waves may cause substantial damage to coastal areas.

12. Which one of the following statements about the 2004 Indian Ocean earthquake and the resulting Tsunami is not correct?

- (a) The earthquake originated due to slipping of about 1,200 km of fault line by 15 m along the subduction zone where the India Plate subducts the Burma plate at the Sunda Trench.
- (b) The resulting tsunami devastated the shores ranging from the coast of Indonesia to the east coast of Africa, some 8,500 km away from the epicenter.
- (c) As per the current estimates, the quake-generated tsunami killed more than 50 lakh people, in addition to unaccounted dead bodies swept out to sea.
- (d) Bangladesh had very few casualties because the quake-affected fault line was in a nearly north-south orientation, the greatest strength of the tsunami waves was in an east-west direction.

47th B.P.S.C. (Pre) 2005

Ans. (c)

Undersea earthquake in the Indian Ocean on 26 December, 2004 produced a tsunami that caused huge natural disaster and over 2 lakh people lost their lives. The earthquake took place in the Indian Ocean off the western coast of Sumatra Island of Indonesia. In regards to the options the details of the tsunami are as follows :

Statement (a) – An estimated 1300 Km. of fault line slipped about 15 metres. The earthquake occurred on the interface of the India and Burma tectonic plates. Thus statement (a) is almost right.

Statement (b) – The waves devastated the shores of Indonesia, Sri Lanka and India as far as Somalia on the east coast of Africa. Hence this statement is also nearly true.

Statement (c) – Approximately 2-3 Lakhs people were estimated to have lost their lives. Thus statement (c) is wrong.

Statement (d) – Bangladesh, which lies at the northern end of the Bay of Bengal had very few casualties. So, statement (d) is true.

13. Which city of Japan had faced tsunami and nuclear radiation recently ?

- (a) Hiroshima
- (b) Tokyo
- (c) Fukushima
- (d) Kobe

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

A major earthquake on 11th March, 2011 with a magnitude (Mw) of 9.0 in Fukushima caused large damage in Japan. Due to this disaster, nuclear power plants were damaged which finally resulted in nuclear radiation.

14. The North-Western Region of the Indian Sub-continent is susceptible to earthquake activity because of –

- (a) Volcanic activity
- (b) plate tectonic activity
- (c) coral formation activity
- (d) All of the above

U.P.P.C.S. (Mains) 2005

Ans. (b)

North-Western region of the Indian Subcontinent is susceptible to earthquake activities because it is situated at the border of plates. This area is convergence region of the Indian Plate and the Eurasian Plate. Coral formation activities and volcanic activities do not occur here.

15. The place on earth's surface which records the seismic wave first, is called :

- (a) Focus
- (b) Epicentre
- (c) Seismosite
- (d) None of the above

U.P.R.O./A.R.O. (Pre) 2017

Ans. (b)